A Resident-Streaming Roofline Model for Compute-in-Memory

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***Abstract*—From CPUs to GPUs, the Roofline model has become a textbook tool for revealing a platform's performance ceiling and how well a workload matches it. Compute-in-memory (CIM), however, falls outside this picture because it violates the core assumption of classical model. Here, we present a resident-streaming Roofline model for CIM, in which streaming-activation and resident-weight throughputs replace compute throughput and memory bandwidth, respectively. This follows from a symmetry we observe: conventional processors pair separated hardware with homogeneous data, whereas CIM pairs unified hardware with heterogeneous data—operands residing in the array and operands streaming through it. Using NeuroSim with literature-extracted parameters, we evaluate various CIM technologies on this Roofline and quantify the demands of mainstream large language models. With similar periphery, the streaming throughputs of these technologies differ by a factor of only about 10^2, whereas their resident throughputs differ by over 10^4. The memory thus decides mainly how much reuse CIM needs, from about 2–10 input vectors per weight load on SRAM and eDRAM to about 4×10^4 on NOR flash, making the reuse a workload offers an explicit criterion for choosing among technologies.**

***Index Terms*—Compute-in-memory, roofline model, streaming intensity, data reuse, large language models.**

I. Introduction

The textbook Roofline model [1] relates hardware capability to workload demand through the Arithmetic Intensity (AI), and exposes the bottleneck arising from a mismatch between soft/hardware. It describes the performance on computation and on data transfer by the compute capability π and the memory bandwidth β, respectively. AI represents the number of operations a workload supports per byte of data transferred; comparing it with the hardware ridge point $\mathrm{AI}^* = \pi/\beta$ distinguishes the memory-bound region from the compute-bound region and provides guidance for optimization [2], [3].

The validity of this model rests on an assumption implicit in the von Neumann architecture: memory and compute are separate units [4]. Fig. 1(a) shows the familiar picture. Storage and computation are carried out by different hardware, and although activations A and weights W play different roles in the algorithm, the hardware treats both simply as operands delivered to the compute units: neither has a distinct identity, and the two are interchangeable.

However, this assumption no longer holds in Compute-in-Memory (CIM), a non-von Neumann architecture [5]. In Fig. 1(b), the CIM macro that stores W also calculates A, so the two hardware units are unified [6]. The ***weights*** take part in computation as ***resident state***, while each new ***activation*** enters through the read path and ***streams across*** the macro, where it is calculated against that state. CIM turns one class of operands, the weights, into stored state at the very site of calculation, while the other class, the activations, flows through the macro as inputs served by that state. The separate hardware units of memory and compute in CPU/GPU disappear, but new roles of streaming and resident operands emerge in CIM.

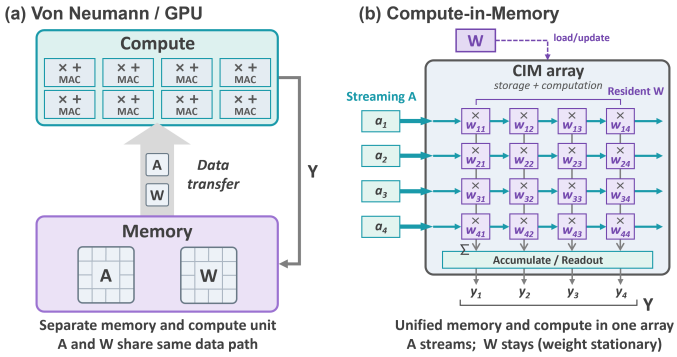


Fig. 1. Hardware schematics: (a) CPU/GPU: separated hardware with homogeneous data; (b) CIM: unified hardware with heterogeneous data.

This change calls for a new Roofline model that rethinks how compute capability and data transfer relate in CIM. The need has two roots. First, the hardware fusion of the CIM architecture dissolves the concept of compute capability π: the capability of a CIM macro to complete multiply-accumulate (MAC) is, in essence, the "transfer rate" of A. Second, the split of operand roles creates two distinct throughputs: weights enter through the write path, which erases, programs, or refreshes the cells, whereas activations enter through the read path, which applies them as inputs and accumulates their products with the weights as outputs. Hence, whereas AI captures the GPU bottleneck as an imbalance between computation and memory access, the CIM bottleneck is the imbalance between these two operand throughputs. Prior studies have recognized this problem [7] and proposed several Roofline analyses for CIM [2], [8]–[14], yet none of them reveals this difference.

Therefore, we propose a resident-streaming Roofline model for CIM. This model separately counts the input bytes Q_S consumed by streaming calculation and the write bytes Q_R that update or reload resident state, and defines the Streaming Intensity $\mathrm{SI} = Q_S/Q_R$ as the streaming demand supported per byte of resident write. Q_S is served by the streaming throughput ρ and Q_R by the resident throughput τ, so the model keeps the form of the classical Roofline, with ρ, τ, and SI taking the places of π, β, and AI. By distinguishing the two operand roles, streaming A and resident W, the model quantifies the imbalance between resident updates and streaming calculation: its ridge $\mathrm{SI}^* = \rho/\tau$ marks where the bottleneck shifts from resident updates to streaming calculation.

Building on this model, we carry out a quantitative analysis from two sides: hardware capability and algorithm demand. On the hardware side, we characterize the streaming and resident throughputs of ten CIM technologies through NeuroSim simulations calibrated with parameters extracted from the literature.

On the algorithm side, we examine the matrix multiplications (GEMM) in mainstream large language models (LLM) and characterize the demands posed by model weights and by key-value (KV) caches. The analysis shows that, with similar periphery, the streaming throughputs of the ten technologies differ by a factor of only about 10^2, whereas their resident throughputs differ by over 10^4. Hence the ridge SI* is limited by the write mechanism of each technology. Comparing the SI of a workload with SI* thus shows which technologies can run it at their streaming ceiling, providing clear guidance for matching CIM technologies to workloads.

II. Proposed Resident-Streaming Roofline

A. Workload Demands and Hardware Throughputs

The classical Roofline describes a workload by its operation count (OPs) and data-transfer volume (Bytes). Following the operand roles in Fig. 1(b), we instead describe a CIM workload through its streaming inputs and resident writes. We count both over an explicit *workload window*, which specifies the work under analysis and the resident state at its start.

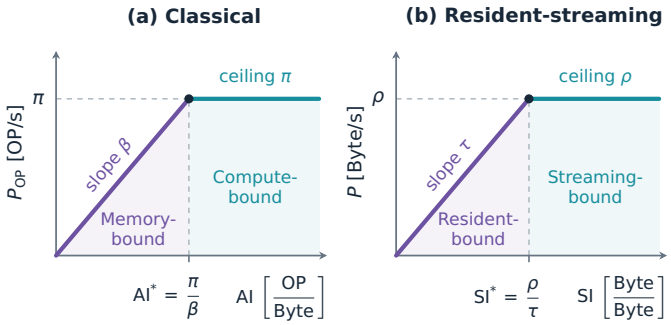
We measure streaming calculation by the logical input bytes evaluated using resident state, with each evaluation (multiplication) producing the specified outputs (partial sum). The streaming demand Q_S is the number of such bytes served at the chosen logical input boundary; it thus represents completed calculation work rather than input/output traffic alone, and it is directly comparable with resident writes. In Fig. 1(b), Q_S counts logical input bytes entering the CIM I/O and broadcast along wordlines (WLs), without counting broadcast copies. The resident demand Q_R is the number of logical bytes written within the window to load or update the resident state that this work requires. An initial load completed before the window is excluded; all updates and reloads within the window are counted. The resident operand may be a static model weight or the dynamic KV cache of earlier tokens.

We characterize two hardware services, streaming-input calculation and resident-state loading, by their throughputs. The streaming throughput ρ is the maximum rate, in logical input bytes per second, at which the hardware completes streaming calculations and produces the specified outputs. The resident throughput τ is the maximum rate at which logical resident bytes become ready for calculation. Both rates refer to the CIM hardware under analysis, with bytes counted at the same interfaces and operand precisions used for Q_S and Q_R, respectively. We define Streaming Intensity and average streaming-input throughput as

$$\mathrm{SI} = \frac{Q_S}{Q_R}, \qquad P = \frac{Q_S}{T} \quad [\mathrm{Byte/s}], \tag{1}$$

where T is the window duration, P is the average streaming-input throughput over the window, and SI is the streaming demand per byte of resident write.

The workload's SI describes the relative demand for streaming calculation and resident writes, while ρ and τ describe how quickly the hardware can perform each. The throughput bound therefore follows from the time required to complete these two kinds of work.



Aspect	GPU (a)	CIM (b)
Decomposition	Hardware resources	Data roles
Workload	OP, Byte	Q_S, Q_R
Service rates	π [OP/s], β [Byte/s]	ρ, τ [Byte/s]
Intensity	AI = OP/Byte	SI = Q_S/Q_R
Vertical axis	P_{OP} = OP/T	P = Q_S/T
Upper bound	$P_{OP} \le \min(\pi, \beta\,\mathrm{AI})$	$P \le \min(\rho, \tau\,\mathrm{SI})$
Left regime	Memory-bound	Resident-bound
Right regime	Compute-bound	Streaming-bound
Ridge	AI* = π/β	SI* = ρ/τ

Fig. 2. Classical GPU [1] and resident-streaming CIM Rooflines with their accounting correspondence. Panels use independent linear scales; curves are upper bounds. OP and Byte denote operation and transferred-byte counts.

B. Theoretical Model and Correspondence

Fig. 2 compares the proposed resident-streaming Roofline with the classical model. Streaming calculation and resident writes require at least Q_S/ρ and Q_R/τ of service time, respectively. The execution must satisfy both constraints, giving

$$P \le \min(\rho, \tau \cdot \mathrm{SI})\,. \tag{2}$$

The two service requirements balance at the hardware ridge $\mathrm{SI}^* = \rho/\tau$. A workload is resident-bound on this hardware when $\mathrm{SI} < \mathrm{SI}^*$ and streaming-bound when $\mathrm{SI} > \mathrm{SI}^*$.

The same bound can be expressed in the familiar GPU-style operation and byte rates. For a window containing O useful operations, the equivalent compute throughput π_{eq} and aggregate bandwidth β_{eq} of CIM are

$$\pi_{\mathrm{eq}} = \frac{O}{Q_S} \cdot \min(\rho, \tau \cdot \mathrm{SI})\,, \tag{3}$$

$$\beta_{\mathrm{eq}} = \left(1 + \frac{Q_R}{Q_S}\right) \cdot \min(\rho, \tau \cdot \mathrm{SI})\,. \tag{4}$$

These are workload-specific projections of the same upper bound: β_{eq} counts the combined logical volume $Q_S + Q_R$, rather than traffic through a GPU memory interface.

Whether a hardware metric is workload-independent depends on the coordinates used to describe it. In GPU, compute

TABLE I
Streaming Intensity after one complete weight load

Weight shape	Reuse count U					
$N \times K$	128	1K	16K	128K	1M	$\to \infty$
128×128	1.0	8.0	128.0	1024.0	8192.0	∞
1024×1024	0.125	1.0	16.0	128.0	1024.0	∞
4096×4096	0.0313	0.250	4.0	32.0	256.0	∞
1024×4096	0.125	1.0	16.0	128.0	1024.0	∞
4096×1024	0.0313	0.250	4.0	32.0	256.0	∞

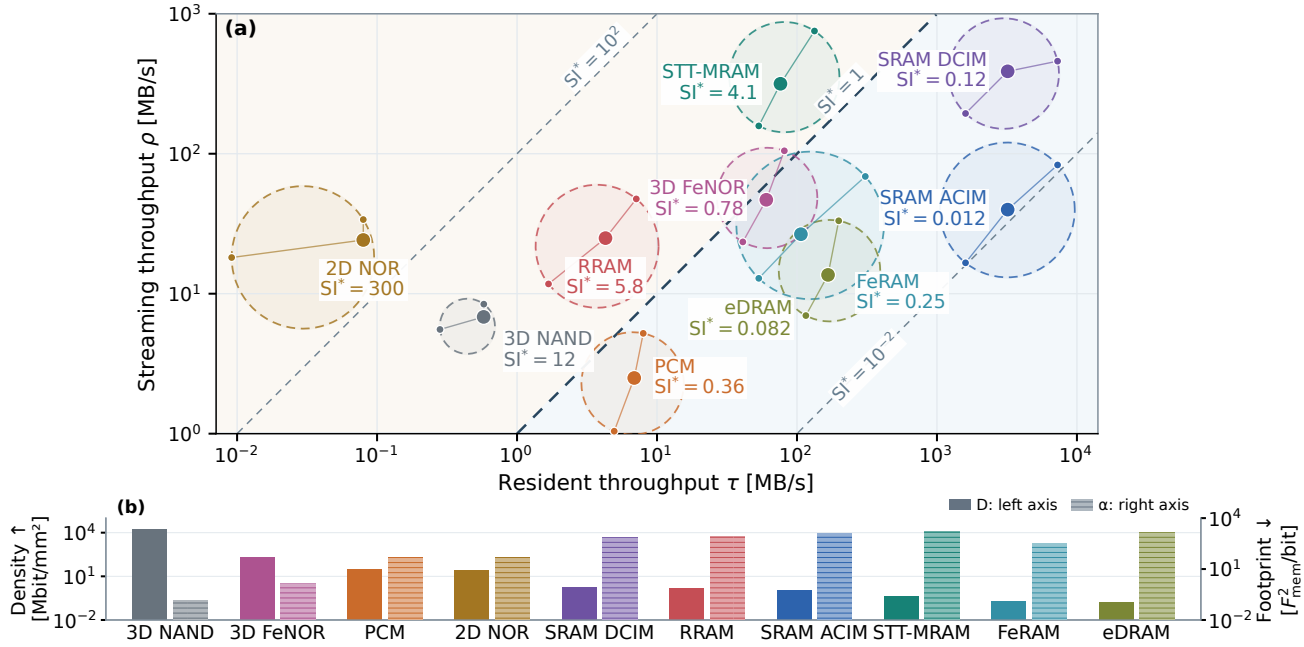


Fig. 3. Throughput and storage compactness of ten CIM technologies. (a) Streaming throughput ρ and resident throughput τ estimated with NeuroSim using literature-extracted parameters. The wider variation in τ drives balance intensities $\mathrm{SI}^* = \rho/\tau$ and reuse requirements. Large dots mark reference configurations; smaller dots and circles show scenario spread. (b) Binary storage density and normalized bit footprint from independent cell/tile geometry references.

and memory units provide distinct services, matched to operation counts and data-transfer volumes. In CIM, storage and computation are combined, but streaming calculation and resident-state loading remain distinct services, matched to streaming-input and resident-write bytes. Each model therefore decomposes hardware services and workload demands along the same dimension—by hardware function for GPUs and by operand role for CIM. This alignment gives each architecture its native capabilities, (π, β) or (ρ, τ), which are fixed for a given configuration. Translating between the two descriptions requires additional information: how many operations the inputs support and how much resident writing is needed. The hardware remains unchanged, but its projected metrics become workload-dependent because the relationship between operations and operand bytes is determined by the workload.

C. Analytical Case Studies

Consider an illustrative 28 nm SRAM ACIM macro serving a logical 128×128 INT8 matrix, with preloaded weights and an assumed input-vector service interval $\Delta_S = 10\,\mathrm{ns}$. Each matrix-vector multiplication consumes 128 input bytes and performs 2×128^2 useful operations, counting two operations per MAC. Thus $Q_R = 0$ and $O/Q_S = 256\,\mathrm{OP/Byte}$, giving

$$\begin{aligned} \beta_{\mathrm{eq}} = \rho &= \frac{128\,\mathrm{Byte}}{10\,\mathrm{ns}} = 12.8\,\mathrm{GB/s}, \\ \pi_{\mathrm{eq}} &= (256\,\mathrm{OP/Byte}) \cdot \rho \approx 3.28\,\mathrm{TOPS}. \end{aligned} \tag{5}$$

The familiar TOPS value is therefore the streaming ceiling expressed in operation units.

We next evaluate SI for GEMM workloads. For a logical matrix $W[N, K]$ with N outputs and K inputs, let U be the number of input vectors served after one complete load, including the first use and allowing reuse across batches or requests. With one byte per input and weight element, $Q_S = UK$, $Q_R = NK$, and $\mathrm{SI} = U/N$. For full-load time T_R and vector service interval Δ_S, the corresponding balance occurs at $U^* = T_R/\Delta_S = N\mathrm{SI}^*$. Table I lists the resulting SI for five matrix shapes and representative reuse counts.

III. Hardware and Workload Characterization

Fig. 3(a) characterizes ten native CIM reference configurations through NeuroSim simulations [15], [16] calibrated with literature-extracted parameters. The inputs come from selected device and macro sources [7], [17]–[45]. Each configuration retains its native dimensions, resources, encoding and update organization under a common peripheral reference policy [46]–[48]. Thus the comparison concerns native services, rather than equal-area chips or uniformly sized arrays. ρ measures the rate of streaming inputs evaluated to the specified outputs, and τ the rate of resident bytes made ready by a complete update. Required erase, program/verify, restore and recovery steps, together with applicable maintenance occupancy, enter the service times. Each configuration has three paired engineering scenarios with fixed organization and resources, giving 30 points. The circles summarize these conditions; they are neither confidence intervals nor sets of freely combinable read and write rates.

Within each scenario, ρ spans about 10^2 across the ten configurations, whereas τ spans more than 10^4. The streaming paths are constrained by input application, sensing and conversion, and/or digital reconstruction under similar peripheral conditions. Resident updates additionally depend on the storage write mechanism, update granularity and actual write-resource organization. For example, reference SRAM ACIM and NOR provide comparable streaming rates of about 40 and 24 MB/s, but resident rates of about 3200 and 0.080 MB/s. Similar

TABLE II
STREAMING INTENSITIES OF REPRESENTATIVE INFERENCE MATRIX STAGES

	Weight projections						Attention					
	QKV (U)			FFN/MoE (U)			Prefill (L)			Decode (L)		
Model	1K	128K	1M	16	1K	16K	1K	8K	64K	1K	8K	64K
Qwen3.5-2B	0.200	25.6	204.8	0.00347	0.222	3.6	6.0	34.0	258.0	10.0	66.0	514.0
Ministral 3 8B	0.167	21.3	170.7	0.00167	0.107	1.7	10.0	66.0	514.0	18.0	130.0	1026.0
Qwen3.6-35B-A3B	0.111	14.2	113.8	0.0130	0.833	13.3	12.0	68.0	516.0	20.0	132.0	1028.0
Hy3 (295B)	0.100	12.8	102.4	0.00477	0.306	4.9	20.0	132.0	1028.0	36.0	260.0	2052.0
Ling-1T (1000B)	0.100	12.8	102.4	0.00326	0.208	3.3	20.0	132.0	1028.0	36.0	260.0	2052.0
MiMo-V2.5-Pro (1020B)	0.0377	4.8	38.6	0.00347	0.222	3.6	35.2	214.4	1648.0	60.8	419.2	3286.4

Entries are SI (one byte/element; $1\mathrm{K} = 1024$, $1\mathrm{M} = 1024^2$). QKV includes any gate but excludes O; FFN/MoE covers one dense FFN or routed expert. Prefill starts with an empty KV cache; decode appends one token to reach visible length L. Ling-1T uses YaRN context extension at 64K.

evaluation capability therefore need not imply a similar cost of establishing state. The wider variation in update service shifts the balance intensity and the reuse needed to sustain streaming calculation.

A memory's ability to retain state compactly is distinct from its ability to update that state quickly. Fig. 3(b) uses independent binary cell/tile geometry references, combining reported geometry with estimates under explicit layout conditions. In particular, NOR uses the GF 28SLPe/SST ESF3 bitcell [49], PCM uses a reported 28 nm FD-SOI 1T1R cell [50], and the FeRAM estimate combines reported 130 nm HZO capacitor data [51] with an assumed access-transistor layout. The last is a conditional model, not a reported complete cell. NAND uses a 500-layer projection of the published lateral geometry [25]. Density and normalized bit footprint express the same geometry in two ways, rather than independently validating it; F_{mem} uses a nominal node or a directional half-pitch as available. These quantities describe neither complete-chip area nor effective INT8 weight density, and a technology's geometry and throughput references need not be the same configuration. Compact storage can make slower-updating memories worth considering for state residency, but this figure alone does not establish an equal-area performance advantage.

On the workload side, Table II selects 72 finite operating points from matrix stages of six fixed models [52]–[57]. These are exact logical counts at one byte per element, obtained from model structure rather than measured execution of complete models. Weight projections and the matrix computations in MLP/FFN or MoE stages follow the one-load, cumulative-use accounting of Section II-C. Composite stages aggregate their actual matrices and count a shared source input once: gate and up projections share an input, while down receives a new input that is counted separately. The load remains part of the window even when the weights are fixed. Consequently, stage structure, input sharing and actual use after loading determine SI; model size or the label "fixed weights" alone does not. Hy3 and Ling-1T, for example, have equal QKV intensity at equal U because their combined projection output widths match, but different single-expert intensities because their expert structures differ.

Attention uses a different resident-state window. Consider the QK and AV matrix stages of one complete causal grouped-query attention (GQA) layer, with H_q query heads, H_{kv} KV heads, native query/key dimension d_{QK} and value dimension d_V, which may differ. Each KV head stores one K/V copy shared by its query group. The streaming inputs are the queries for QK and the post-softmax attention coefficients for AV; historical K/V remains resident. Prefill builds the cache from empty to visible length L, giving

$$Q_S = LH_q[d_{\mathrm{QK}}+(L+1)/2], \quad Q_R = LH_{\mathrm{kv}}(d_{\mathrm{QK}}+d_V). \tag{6}$$

The triangular term accumulates the coefficients over all causal prefixes, while each token's K/V is written only once. Decode instead starts with $L-1$ tokens of KV and appends one token before evaluating against length L:

$$Q_S = H_q(d_{\mathrm{QK}} + L), \quad Q_R = H_{\mathrm{kv}}(d_{\mathrm{QK}} + d_V). \tag{7}$$

The ratio of the two counts in each window gives its SI. Decode writes only the new K/V but evaluates against the full visible history; the existing cache is not written again in this window. Dynamic state can therefore support high intensity when each increment enables substantial streaming work. Even within one model, the stage and operating window determine the demand to be matched to hardware services.

IV. WORKLOAD MATCHING AND DESIGN IMPLICATIONS

Fig. 4 compares six operating points of Qwen3.6-35B-A3B from Table II with the ten native hardware balance intensities; each range covers the three paired scenarios. This is a reference comparison of logical stage demands and native service ratios. A deployment must align their counting boundaries for input sharing, replay and the corresponding update organization before inferring its limiting service. Relative to each workload line, hardware with $\mathrm{SI}^* < \mathrm{SI}$ lies on the streaming side, and hardware with $\mathrm{SI}^* > \mathrm{SI}$ on the resident side; each line defines its own boundary.

For the same MoE expert, increasing U from 16 to 1K leaves the complete weight load Q_R unchanged and raises SI from about 0.0130 to 0.833. All three PCM balance intensities, spanning 0.211–0.653, lie between these demands. The reference comparison therefore moves from the resident side to the streaming side in every PCM scenario. Extending effective use of resident state can change the limiting service

without changing the hardware. Achieving that use in a routed model still depends on which inputs reach the expert before its weights are replaced.

Near balance, the particular service conditions matter. Prefill at $L = 1\text{K}$ has $\text{SI} \approx 12.008$, just above the NAND reference $\text{SI}^* \approx 11.830$, but below its other two scenarios at about 14.548 and 19.722. The reference point is on the streaming side while both alternatives remain on the resident side. One reference point thus cannot represent the limiting service across the full set of engineering conditions.

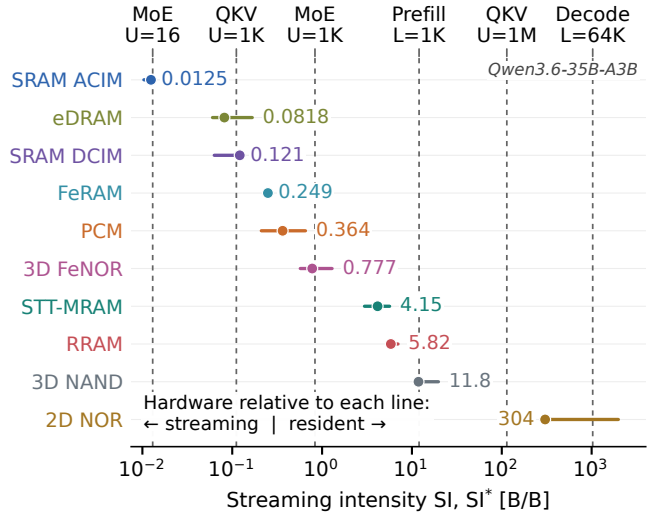


Fig. 4. CIM–workload matching. Dots and ranges show hardware balance intensities SI^*; dashed lines show workload SI. Technologies left of each line are streaming-bound; those to its right are resident-bound.

A lower balance intensity alone does not imply a higher performance ceiling: it can result from a larger τ or a smaller ρ. Reference PCM has $\text{SI}^* \approx 0.364$, below STT-MRAM's 4.15, yet their native streaming rates are about 2.50 and 316 MB/s, respectively. These ratios describe how the two services balance, not a model-speed ranking. Fig. 3(a) therefore remains necessary to retain absolute service rates when interpreting Fig. 4. Its companion geometry panel adds the capacity potential of retaining state, which neither service rate captures.

Once demand and capability boundaries are aligned, a resident-limited workload calls for faster effective updates or greater SI through actual state retention and fewer reloads. Raising ρ alone cannot remove the current τSI limit. Conversely, once streaming service limits the work, faster writing alone cannot raise the streaming ceiling; evaluation service must improve. A target intensity gives the update-service budget $\tau \geq \rho/\text{SI}_{\text{target}}$ needed to avoid lowering that ceiling, although satisfying this condition does not guarantee reaching it.

Storage compactness can support these choices by allowing more state to remain resident. Translating cell density into fewer reloads, however, requires a deployment capacity and mapping analysis; Fig. 3(b) alone establishes neither whole-model fit nor an attainable U. The update organization must also follow the workload window: fixed weights here incur a complete matrix load, whereas decode appends new KV to existing state. The KV service budget must therefore reflect the actual append transactions; the full-load reuse threshold does not transfer unchanged.

V. Conclusion

Outline. Close with the role-based connection between in-window resident updates and streaming evaluation established by (2). Use Fig. 4 to emphasize the practical outcome: hardware service asymmetry becomes an explicit reuse requirement for residency and update decisions.

BODY TEXT RESERVED

Final conclusion: one short paragraph.

Writing round 1: prose not yet drafted.

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